

A Technical Introduction to Encoder decoder architecture

Section 1

In words, it means that each token can pay attention to itself, and every token before it, but not any after it. A non-masked attention module can be thought of as a masked attention module where the mask has all entries zero. As an example of an uncommon use of mask matrix, the XLNet considers all masks of the form $P \text{ M causal } P - 1$ $\{\text{displaystyle PM}_{\{\text{causal}\}}P^{-1}\}$, where P $\{\text{displaystyle P}\}$ is a random permutation matrix.

Section 2

Sub-quadratic transformers Training transformer-based architectures can be expensive, especially for long inputs. Many methods have been developed to attempt to address the issue. In the image domain, Swin transformer is an efficient architecture that performs attention inside shifting windows. In the audio domain, SepTr decouples the attention in time and frequency domains. Long Range Arena (2020) is a standard benchmark for comparing the behavior of transformer architectures over long inputs.

Section 3

Un-embedding An un-embedding layer is almost the reverse of an embedding layer. Whereas an embedding layer converts a token identifier into a vector, an un-embedding layer converts a vector into a probability distribution over tokens.

Section 4

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/* first sublayer */ z_d_copy ← copy(z_d) for each t in 1:length(z_d) do z_d[t] ← layer.layer_norm(z_d[t]) z_d ← layer.masked_multihead_attention(z_d, z_d, z_d) for each t in 1:length(z_d) do z_d[t] ← z_d[t] + z_d_copy[t]
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Section 5

FlashAttention FlashAttention is an algorithm that implements the transformer attention mechanism efficiently on a GPU. It is a communication-avoiding algorithm that performs matrix multiplications in blocks, such that each block fits within the cache of a GPU, and by careful management of the blocks it minimizes data copying between GPU caches (as data movement is slow). See the page on softmax for details. An improved version, FlashAttention-2, was developed to cater to the rising demand for language models capable of handling longer context lengths. It offers enhancements in work partitioning and parallelism, enabling it to achieve up to 230 TFLOPs/s on A100 GPUs (FP16/BF16), a 2x speed increase over the original FlashAttention. Key advancements in FlashAttention-2 include the reduction of non-matmul FLOPs, improved parallelism over the sequence length

dimension, better work partitioning between GPU warps, and added support for head dimensions up to 256 and multi-query attention (MQA) and grouped-query attention (GQA). Benchmarks revealed FlashAttention-2 to be up to 2x faster than FlashAttention and up to 9x faster than a standard attention implementation in PyTorch. Future developments include optimization for new hardware like H100 GPUs and new data types like FP8. FlashAttention-4 focuses on pipelining to increase instruction throughput, and was developed to perform particularly well on Blackwell GPUs.

Section 6

Methods for stabilizing training The plain transformer architecture had difficulty in converging. In the original paper, the authors recommended using learning rate warmup. That is, the learning rate should linearly scale up from 0 to maximal value for the first part of the training (usually recommended to be 2% of the total number of training steps), before decaying again. A 2020 paper found that using layer normalization before (instead of after) multihead attention and feedforward layers stabilizes training, not requiring learning rate warmup. This is the "pre-LN Transformer" and is more commonly used, compared to the original "post-LN Transformer".

Section 7

The idea of encoder–decoder sequence transduction had been developed in the early 2010s; commonly cited as the originators that produced seq2seq are two concurrently published papers from 2014. A 380M-parameter model for machine translation uses two long short-term memories (LSTM). Its architecture consists of two parts. The encoder is an LSTM that takes in a sequence of tokens and turns it into a vector. The decoder is another LSTM that converts the vector into a sequence of tokens. Similarly, another 130M-parameter model used gated recurrent units (GRU) instead of LSTM. Later research showed that GRUs are neither better nor worse than LSTMs for seq2seq. These early seq2seq models had no attention mechanism, and the state vector is accessible only after the last word of the source text was processed. Although in theory such a vector retains the information about the whole original sentence, in practice the information is poorly preserved. This is because the input is processed sequentially by one recurrent network into a fixed-size output vector, which is then processed by another recurrent network into an output. If the input is long, then the output vector would not be able to contain all relevant information, degrading the output. As evidence, reversing the input sentence improved seq2seq translation. The RNN search model introduced an attention mechanism to seq2seq for machine translation to solve the bottleneck problem (of the fixed-size output vector), allowing the model to process long-distance dependencies more easily. The name is because it "emulates searching through a source sentence during

decoding a translation". The relative performances were compared between global (that of RNN search) and local (sliding window) attention model architectures for machine translation, finding that mixed attention had higher quality than global attention, while local attention reduced translation time. In 2016, Google Translate was revamped to Google Neural Machine Translation, which replaced the previous model based on statistical machine translation. The new model was a seq2seq model where the encoder and the decoder were both 8 layers of bidirectional LSTM. It took nine months to develop, and it outperformed the statistical approach, which took ten years to develop.

Section 8

The original version of the transformer architecture was proposed in the 2017 paper "Attention Is All You Need" by researchers at Google. The predecessors of transformers were developed as an improvement over previous architectures for machine translation, but have found many applications since. They are used in large-scale natural language processing, computer vision (vision transformers), reinforcement learning, audio, multimodal learning, robotics,, playing chess and at disaster responses. It has also led to the development of pre-trained systems, such as generative pre-trained transformers (GPTs) and BERT (bidirectional encoder representations from transformers).

Section 9

The un-embedding layer is a linear-softmax layer: $\text{UnEmbed}(x) = \text{softmax}(xW + b)$ $\{\text{UnEmbed}(x) = \text{softmax}(xW + b)\}$ The matrix has shape $(d_{\text{emb}}, |V|)$ $\{(d_{\text{emb}}, |V|)\}$. Some architectures use the transpose of the embedding matrix M $\{M\}$ as the un-embedding matrix W $\{W\}$ in order to avoid needing double the amount of embedding-related parameters and to avoid divergence during training. This practice is called weight tying.